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Ventilator Management on ECMO

By this point you have probably developed a healthy skepticism for the ventilator. While remaining an undeniable cornerstone in the support of critically ill patients, we have gone into the myriad of limitations the ventilator has in severe respiratory failure, when compliance worsens and it becomes progressively difficult to oxygenate/ventilate.

Now we will progress to some principles for management of the ventilator while on extracorporeal membrane oxygenation (ECMO). It should be noted that this is one of the areas where there is probably the most variation in practice patterns, with some practitioners being very conservative with ventilator weaning, and others being very aggressive. Rather than advocating one approach over the other, we will try to focus this chapter on principles that can be applied in order to minimize the damaging effects of the ventilator for patients on ECMO. This will not be a comprehensive overview of all strategies of mechanical ventilation, but rather a very focused review of principles that are commonly applied to patients on ECMO.

We will focus this discussion on patients with primary respiratory failure, however the principles discussed will apply to all types of support.

COMMON VENTILATOR SETTINGS IN ECMO SUPPORT

Imagine you are caring for a patient who was just initiated on ECMO. As support was initiated, they had a phenomenal response – their saturations increased from 77% to 100%, their heartrate decreased from 120 to 98, and their blood pressure improved from 89/55 to 105/70.

Your eyes now turn to the ventilator. It reads as follows:

> Mode AC/VC. Tidal Volume 500. 100% FiO_2. PEEP 14. Rate 34. Peak pressures measured at 42.

Just as you are pondering the ventilator and the settings that you can transition this patient to, the respiratory therapist asks, “Do you want me to change the patient to our typical vent settings?”

She then makes the following changes to the ventilator.

> Mode AC/PC. Inspiratory pressure 12. 50% FiO_2. PEEP 10. Rate 26. Tidal volume measured at 145 mL.

What is the difference between these two settings? What is the rationale behind these changes?

MODE OF VENTILATION: SPONTANEOUS VERSUS CONTROLLED VENTILATION

The mechanism of the ventilator is relatively straightforward – it pushes a set amount of air into the lungs. As such, you can imagine that there really are three primary variables that effect how this air is delivered: what initiates the breath (referred to as **trigger**), what is delivered with each breath (referred to as **limit**), and what ends the breath (referred to as **cycle**).

If the ventilator is delivering a set number of breaths regardless of whether the patient is initiating the breath or not, this is controlled ventilation. Contrast this to spontaneous ventilation, where every breath that is taken has to be initiated by the patient (Figs. 19.1 and 19.2).

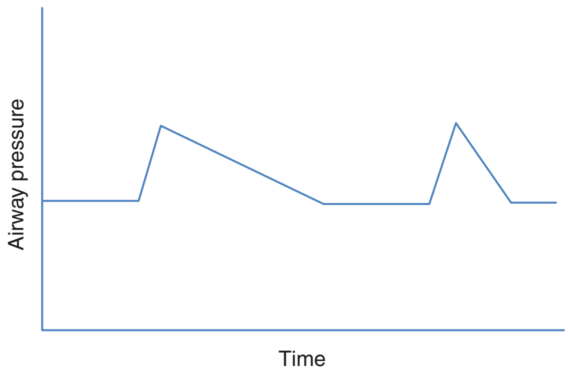

FIG. 19.1 Airway pressures in spontaneous ventilation

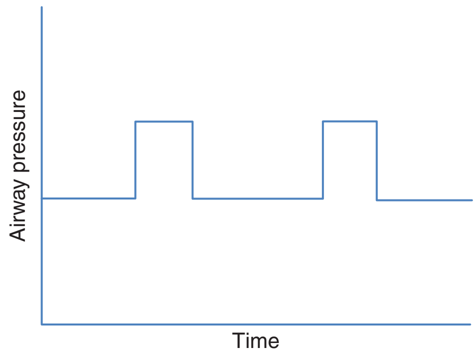

FIG. 19.2 Airway pressure in controlled ventilation

The other difference between these two modes is how the breath ends. In spontaneous mode, as you can see earlier, the breath can be long or short and is only stopped when a specific flow has been obtained. Compare this to controlled ventilation, in which the breath is stopped after a set amount of time. This may result in more variation in the size of the breath for spontaneous ventilation, which is similar to how we breathe without the ventilator, but may or may not be well tolerated in the setting of severe lung dysfunction.

In spontaneous ventilation, a certain amount of positive end-expiratory pressure (PEEP) is set followed by a set amount of pressure that is delivered above PEEP with every breath. The volume of breath that is generated is dependent on the flow that the patient is able to generate, based on the strength of the diaphragm.

This reliance on the diaphragm forms the main advantages and disadvantages of spontaneous ventilation. On one hand, this mode can condition the diaphragm, employing it in every breath similar to breathing without the ventilator. However, if there is severe decompensation, this may lead to overexertion and weakness.

How do you make the distinction between selecting spontaneous versus controlled ventilation? This is a challenging question to answer due to the wide range of approaches taken by various practitioners, as well as the varied response that individual patients can have. As a general rule, controlled ventilation may be needed early in the course, with the aim of allowing for spontaneous ventilation as the patient continues to wake up and recover. This can be over the course of several days or several hours, depending on how the patient is responding to weaning. Wean as much as is tolerated clinically.

What happens when spontaneous breathing is not feasible clinically? Let's go over a couple of considerations for controlled ventilation.

Controlled Ventilation: Volume Versus Pressure Control

The two main modes for controlled ventilation are pressure control and volume control. The difference between the two is as follows:

Volume control: delivers a set amount of **volume** with each controlled breath (flow limited/volume cycled)

Pressure control: delivers a set amount of **pressure** with each controlled breath (pressure limited/time cycled)

Minimizing both pressure and volume delivered is ideal, since as you remember, both can cause damage to the lungs in the form of barotrauma (pressure) and volutrauma (volume). The challenge comes when compliance decreases, as occurs when lungs get progressively stiff due to edema, acute respiratory distress syndrome (ARDS), pneumonia, hemorrhage, or any of the host of conditions that are associated with severe cardiac/respiratory failure. In this case, you are left with either tolerating higher pressures or lower tidal volumes, as denoted below:

$$\downarrow \text{Compliance} = \frac{V_T}{\text{PIP} - \text{PEEP}} \quad \begin{cases} \downarrow V_T \\ \uparrow P_{IP} \end{cases}$$

This is the tradeoff that often has to occur with worsening respiratory failure that leads to a failure to ventilate: either you have to tolerate higher CO_2 levels due to lower tidal volumes or you have to tolerate higher airway pressures in order to achieve a target tidal volume.

ECMO obviates the need for this choice. Since the circuit is able to remove CO_2 with great efficiency, you can significantly lower both the pressure and volume delivered with each breath and afford better lung protection from the damaging effects of barotrauma and volutrauma.

Either mode is acceptable as long as you are providing lung-protective ventilation.

Lung Protective Variable #1: Tidal Volume

Tidal volume is one of the mechanical ventilation parameters with the most well-established association to improved outcomes. We know that high tidal volumes are damaging to the lung parenchyma and that this damage has been linked to a higher mortality.

The generally accepted target is 6 mL per kg of ideal body weight. When compared to higher tidal volumes, tidal volumes of less that 6 mL/kg have been associated with improved mortality. What about 5 mL/kg? 4 mL/kg? 2 mL/kg?

6 mL/kg is usually the target because lower tidal volumes often leads to difficulty in removing CO_2, due to lower minute ventilation (tidal volume × breaths per minute). However, CO_2 removal is less of an issue in ECMO, allowing for lower tidal volumes if desired.

Thus, many practitioners aim for tidal volumes on ECMO less than **4 mL/kg**. This is sometimes referred to as "ultra–lung-protective ventilation." There is some evidence to support this strategy, as 4 mL/kg was the lower limit of normal in many trials.

Lung Protective Variable #2: Driving Pressure

Let's introduce this concept of driving pressure. Driving pressure is defined as follows:

Driving pressure = Plateau Pressure − PEEP

Driving pressure is not discussed as much as tidal volume, but actually has a higher association with improved outcomes.

Before we discuss driving pressure, let's examine what is meant by plateau pressure.

Recall in our discussion of flow dynamics, we used the example of two balloons connected by a plastic tube. We noted that the flow of air between the balloons is determined by the pressure difference between these balloons as well as the resistance of the tube. We used this example to illustrate the limits to blood flow through the ECMO circuit. Now let's apply this dynamic to the flow of air from the ventilator to the lungs (Fig. 19.3).

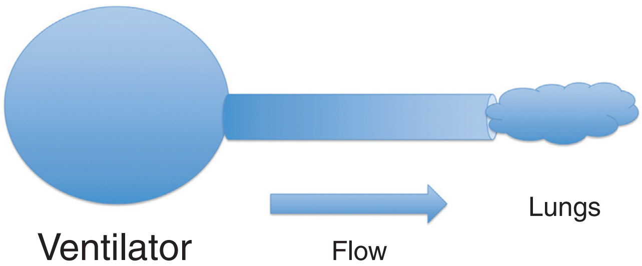

FIG. 19.3 Airway pressure and ventilation

In this case, there is a specific amount of pressure that has to be delivered to inflate the lungs based on the compliance of the lungs, but also a specific amount of pressure that has to be delivered to overcome the resistance of everything that connects the ventilator and the alveoli (such as the ventilator circuit, the endotracheal tube, the trachea, and the bronchi).

These two pressures can be differentiated by doing an inspiratory breath hold, which will measure the pressure required to distend the alveoli only as the resistance of the circuit is already overcome, known as the plateau pressure. Let's compare differences in plateau pressures for two patients with an elevated peak pressure. If there is minimal resistance in the connection, then the plateau pressure will be close to the peak pressure (blue line). On the other hand, if there is high resistance in the connection (due to kinking of the endotracheal tube, mucous plugging in the trachea, or spasm in the bronchi as examples), then the plateau pressure will be much lower than the peak pressure, as noted by the red line (Fig. 19.4).

The difference between the PEEP and the plateau pressure, as measured by an inspiratory breath hold, is the driving pressure. As a general rule, the lower the driving pressure the better. This can be accomplished by lowering the plateau pressure (through lower inspiratory pressures

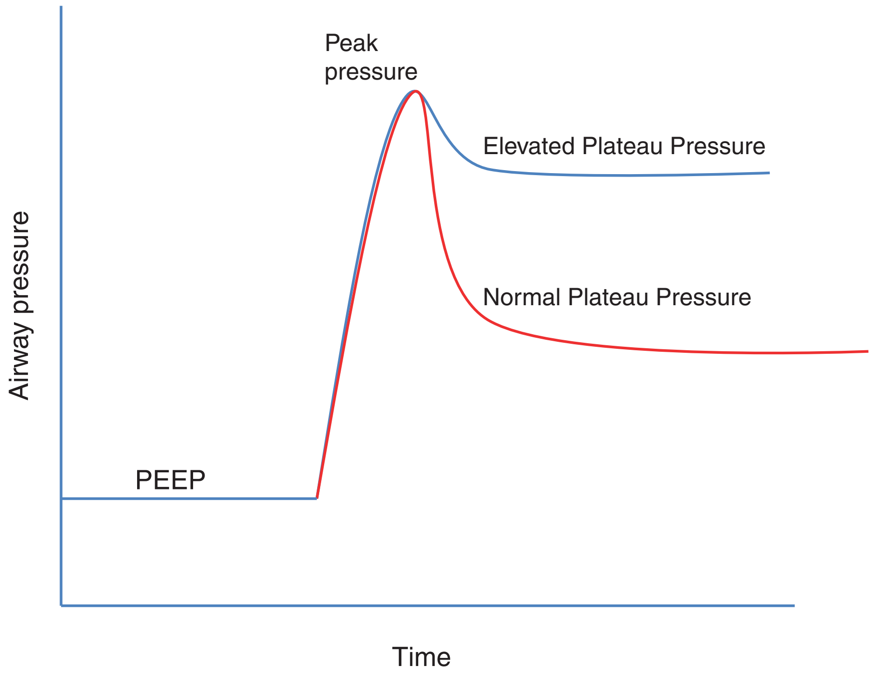

FIG. 19.4 Comparison of elevated plateau pressure (blue line) v. normal plateau pressure (red line). PEEP, Positive end-expiratory pressure

on pressure control or lower tidal volumes on volume control) or by increasing the PEEP. A driving pressure of **less than 10–15 cm H_2O** is a good target.

If increasing PEEP does not lead to a higher plateau pressure, this indicates lung recruitment and has been associated with a better outcome. Let's now discuss PEEP.

POSITIVE END-EXPIRATORY PRESSURE TARGETS DURING ECMO

PEEP is the minimum amount of positive pressure maintained by the ventilator, defined as the pressure at the end of expiration, as illustrated in Fig. 19.5. At a basic level, there are several benefits to maintaining some PEEP:

1. Prevents total alveolar collapse allowing for a specified number of alveoli to remain open
2. Facilitates improved oxygenation by raising mean airway pressure due to higher time spent in the expiratory phase
3. Allows for convective ventilation by keeping more alveoli open and participating in gas exchange
4. Improves hemodynamics by decreasing cardiac afterload

Too little PEEP can be detrimental; however, excessive PEEP can have an adverse effect by a host of mechanisms including:

1. Overdistension of alveoli, increasing the proportion of alveoli with dead space ventilation
2. Eventually raising plateau pressure, leading to barotrauma
3. Decreasing preload to right heart
4. Increasing right ventricular afterload

Finding the right amount of PEEP is key. One way to assess for ideal PEEP is to observe the effect of adjustments in PEEP to the patient. If increasing PEEP leads to improved hemodynamics,

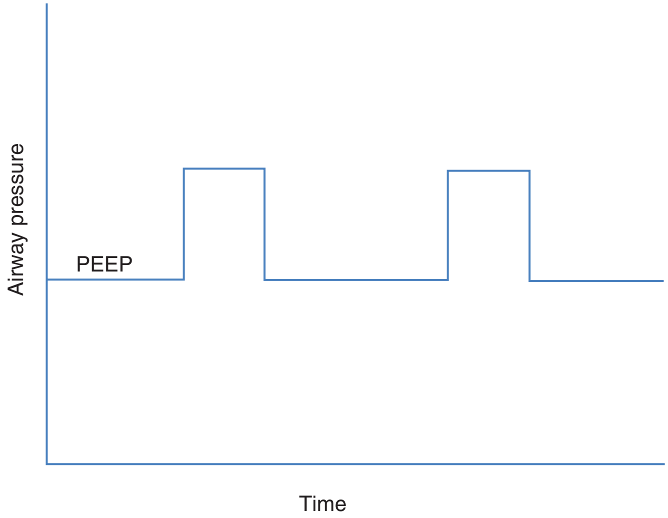

FIG. 19.5 PEEP and overall airway pressure. PEEP, Positive end-expiratory pressure

no change in plateau pressure, and better oxygenation, then this is an indicator that this change is well tolerated.

By convention, PEEP of **8–15 cm H_2O** is a reasonable starting target for patients on ECMO. Adjustments should be made thereafter based on the response of the patient.

LIMITING FiO_2 FROM THE VENTILATOR

Limiting FiO_2 from the ventilator is essential to prevent damage to the patient, but doing so takes a great deal of discipline. This is because patients on ECMO are often hypoxic, and it is difficult to see the harmful effects of high oxygen levels. In reality, high levels of FiO_2 can make it very difficult for lungs to improve. Let's explore the harm that can come from prolonged exposure to 100% FiO_2.

As we discussed in Chapter 5, when we normally breathe room air, only around 21% of the air we are breathing is oxygen. The rest is predominantly nitrogen (N_2), for which there does not exist the same diffusion gradient between the air in the alveoli and the blood as exists for O_2. As a result, the nitrogen that enters the alveoli remains in the alveoli for the most part. This nitrogen plays a key role in keeping the alveoli open.

Oxygen, on the other hand, is a key component of metabolism; therefore the blood that is delivered to the lungs will have a lower partial pressure of oxygen. This drives the diffusion gradient that allows oxygen to enter the blood from the air (Fig. 19.6).

In the case of room air, oxygen is able to enter the blood and the alveoli are kept open due to the nitrogen present. However, as the FiO_2 increases to 100%, then all of the oxygen that enters the alveoli diffuses into the blood and there is nothing to keep the alveoli open. This leads to atelectasis and furthers damage (Fig. 19.7).

This is also combined with diminishing returns from escalating amounts of oxygen. Recall that as shunt fraction increases, the amount of oxygen administered does not improve the overall oxygenation of the blood.

As a general rule, try to keep the FiO_2 as low as possible, with the caveat that you may be limited by hypoxia. Hypoxia may sometimes have to be tolerated. Maintaining FiO_2 less than 60% allows for some nitrogen and can promote lung improvement and recruitment.

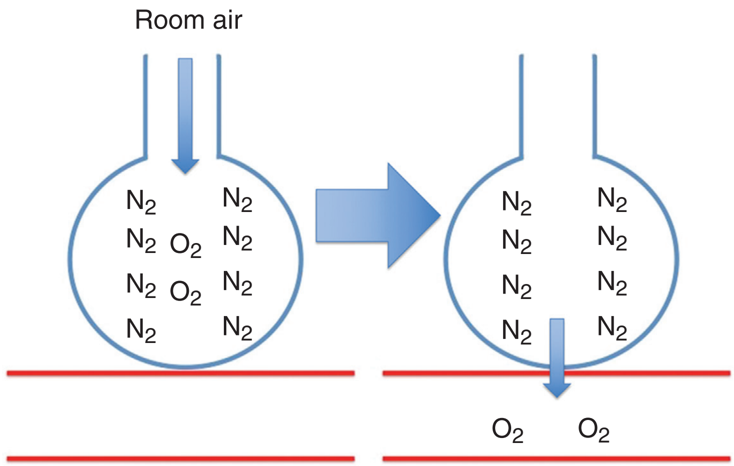

FIG. 19.6 The role of N_2 in maintaining open alveoli in room air

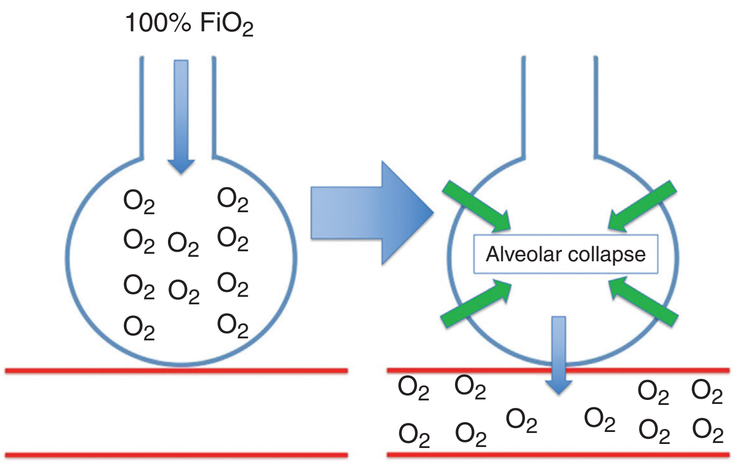

FIG. 19.7 Alveolar collapse from 100% FiO_2

A NOTE ON HYPOXIA ON ECMO

If you walk by the bedside of a patient on ECMO, you may notice an oxygen saturation in the 80 s, 70 s, or lower. The reason for this may be that lung-protective ventilation is being maintained, and high levels of FiO_2 and aggressive maneuvers to force recruitment have been deemed harmful or, at the very least, less likely to be of benefit.

This is an important distinction – hypoxia on ECMO is not *targeted* but sometimes is *tolerated*, when the alternative involves a strategy that may be harmful to the patient. If the patient is not showing adverse effects of decreased DO_2 (for example, lactic acidosis, decreased urine output, or decreased mental status), the benefits of lung protection may outweigh the risks of hypoxia.

CONTROLLING THE RESPIRATORY RATE

Respiratory rate is the second determinant of minute ventilation along with tidal volume. Since tidal volume is much more closely associated with adverse outcomes, improvement in minute ventilation/CO_2 clearance is usually accomplished through increases in respiratory rate in patients not supported by ECMO.

However, at a certain point, respiratory rate can be so high as to minimize the effectiveness. At rates greater than 30–34, there is limited chest recoil between breaths, and a higher proportion of the breath that is only moving through anatomic dead space, such as the trachea and bronchi. Think of a dog that is panting – high respiratory rate, but a very limited exchange of gases.

Now consider the patient on ECMO. This patient is much less dependent on minute ventilation for CO_2 clearance. Decreasing the respiratory rate has several positive effects to include:

1. Better chest recoil and ventilation with the breaths delivered
2. Less chance of incomplete exhalation leading to auto-PEEP, with potential hemodynamic consequences
3. Less overall strain on the respiratory apparatus

You want to allow enough of a rate to keep the patient comfortable, and to allow for some exchange of gas throughout the lung. In general, the accepted number is somewhere between **8 and 20 breaths per minute**.

Be warned, however, that respiratory rate may be very difficult to actually control, no matter what is set to the ventilator. This is because there may exist an impetus to breathe quickly despite the $PaCO_2$ and PaO_2 levels that the patient is experiencing.

The drive to breathe is a complex process that is driven by multiple pathways and receptors throughout the central nervous system to include central chemoreceptors located in the medulla that drive respiratory rate based on CO_2/O_2 levels in the blood, the pneumotaxic center in the pons that drives normal respiratory drive, and irritant/J receptors that exist at the level of the lungs. As part of our primitive reflexes, these irritant/J receptors can stimulate fast breathing in the setting of lung inflammation, as is common in patients with lung/heart failure.

The take-home point being that patients may breathe fast as a function of their underlying disease process. This high respiratory rate may be tolerated or mitigated through higher sedation, weighing the relative risks and benefits.

EXTUBATION ON ECMO: WHY USE THE VENTILATOR AT ALL?

It is now four days later for our patient that we met at the beginning of this chapter. We have decreased driving pressure and tidal volumes in such a manner to allow for ultra–lung-protective ventilation. FiO_2 is now down to 40%. PEEP has been optimized and remains at 8. Respiratory rate was taken down to a reasonable range of 16 before transitioning the patient to spontaneous mode, which he is tolerating well.

Should we continue weaning? Do we need the ventilator at all?

Extubating and taking the patient off the ventilator is an intriguing possibility. The best way to avoid complications such as ventilator-associated pneumonia and ventilator-induced lung injury is to take the ventilator out of the room!

Normally, ventilator-weaning parameters are well established in an ICU to facilitate timely cessation of mechanical ventilation, with the intended consequence of decreasing ventilator times and length of stay. These parameters should be considered for patients on ECMO but should be contextualized – the severity and pathology of cardiac/respiratory failure should be considered as well as the fact that the ECMO circuit may make some parameters such as $PaCO_2$ and PaO_2 look falsely reassuring. Prior to consideration of extubation on ECMO, the following three conditions must be met.

Condition For Extubation #1: Ventilator Is Not Needed to Maintain Respiratory Parameters

"Why do we even need the ventilator on ECMO? Isn't the circuit was providing for ventilation/oxygenation?" you may be asking yourself.

Recall our discussion on the limits of the membrane oxygenator in Chapter 10 and our illustration of the tennis court. We mentioned how the surface area for gas exchange is almost 40 times larger for the lungs and 1/10 the thickness. This means that even if lungs are severely compromised, they may be needed to maintain oxygenation/ventilation while on ECMO.

The main change that will be lost during extubation will be the loss of PEEP. This PEEP may lead to significant derecruitment and loss of oxygenation on the part of the native lungs. Whenever possible, efforts to provide safe, lung-protective recruitment (intermittent positive pressure ventilation, respiratory treatments, incentive spirometry, use of tracheal speaking valve, upright/seated positioning, and even noninvasive positive pressure ventilation in some cases) should be considered for patients who are extubated, especially right after extubation.

Condition For Extubation #2: Ventilator Is Not Needed to Maintain Hemodynamic Parameters

PEEP increases the overall intrathoracic pressure, which can impact hemodynamics and the overall function of the heart. In some cases, PEEP can worsen the function of the heart, where in other cases PEEP can improve cardiac function.

In either case, the hemodynamic effects of the loss of PEEP should be considered prior to proceeding to extubation.

Condition For Extubation #3: Ventilator Is Not Needed to Maintain Airway

Besides optimization of oxygenation/ventilation, the other main advantage of the ventilator is the protection of the airway either due to loss of mental status or excessive risk of aspiration. Loss of mental status is a way that the clinical presentation of patients on ECMO can be deceiving. Because the circuit is providing oxygenation and ventilation, patients may breathe comfortably, with no signs of distress. However, this does not mean that they will be able to robustly protect their airway when the endotracheal tube is removed. This places the patient at risk for aspiration, which can be disastrous in the setting of an already present cardiac/respiratory failure.

Going along with this concept is ensuring protection from anything that could be aspirated. Considerations here include heavy oral/respiratory secretions, vomiting/high gastric output, or blood (hematemesis, epistaxis, hemoptysis).

Finally, when considering airway protection, consider the anticipated clinical course. For example, if trips to the OR/catheterization lab are anticipated, mental status has been waxing/waning due, or evolving infiltrates on the chest X-ray indicate that respiratory function may be worsening, then leaving the endotracheal tube in place may be prudent in the near term.

If all conditions are met, extubation has many advantages to include reduced sedation requirements, decreased delirium, facilitation of rehabilitation, and an improved ability for the patient to communicate with family and loved ones.

PUTTING IT TOGETHER

We discussed the principles behind the settings commonly seen in ECMO. To review, the following are the goals that we can attempt to meet during mechanical ventilation.

Goals during mechanical ventilation
Spontaneous mode when clinically tolerated
FiO_2 as low as possible (ideally <60%)
PEEP: 8–12 cm H_2O
Tidal volume in controlled mode: <4 mL/kg
Driving pressure in controlled mode: <15 cm H_20
Respiratory rate: 8–20 breaths/min
Extubation when not limited by hemodynamics, airway protection, or derecruitment

The mechanical ventilation strategy on ECMO is complex and can take a lifetime to master. Rather than providing a comprehensive approach to mechanical ventilation, this chapter aims to help give reasoning and principles behind the mechanical ventilation settings most commonly used while supporting patients on ECMO, both due to cardiac and respiratory failure.

Hopefully, the discussion can be a starting point to introduce concepts, with the understanding that individual patients will have unique needs that will dictate the ideal strategy. If your strategy is allowing for minimal damage from the ventilator and facilitating heart/lung recovery, then this is a good indicator that you are heading the right direction.

SUGGESTED READING

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